A Sustainable Hybrid QKD–PQC Security Architecture for Quantum-Resilient Residential IoT

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***Abstract*—The rapid proliferation of Internet of Things (IoT) technologies across residential environments creates a security problem that is fundamentally intertwined with computing sustainability. Constrained smart home nodes operate under strict energy, memory, computational, and bandwidth bounds while facing the threat posed by quantum cryptanalysis via Shor's algorithm.**

This paper proposes a sustainable hybrid security architecture for quantum-resilient residential IoT. The architecture introduces an edge-centric Quantum-Safe Home Security Gateway (QHSG) that structurally isolates security-intensive key-management operations from high-volume application data streams. Last-mile access-node Quantum Key Distribution (QKD) key material is incorporated where optical infrastructure is available, while standardized Post-Quantum Cryptography (PQC) mechanisms provide a deployable security path over conventional networks. Bulk application data is secured using authenticated symmetric encryption.

The principal contribution is a security–sustainability framework that formalizes computational energy, radio communication overhead, 6LoWPAN fragmentation penalties, and adaptive key-refresh intervals as explicit design parameters. We validate the architecture through a benchmark-calibrated empirical simulation modeling ARM Cortex-M4 leaf endpoints across network scales from $N = 10$ to $N = 500$ devices, augmented by protocol-level validation of the 6LoWPAN fragmentation model in COOJA and by symbolic formal verification of the key-lifecycle and refresh-policy security invariants in ProVerif. The results, obtained at an IEEE 802.15.4 physical rate of 250 kbps with radio constants anchored to device datasheets, demonstrate that direct leaf-node PQC deployment increases endpoint key-negotiation energy by 108.4% over classical handshakes due to transmission and fragmentation of large lattice keys. In contrast, the proposed QHSG-mediated hybrid architecture offloads asymmetric encapsulation to the gateway, reducing endpoint key-management energy by 96.8% relative to standalone PQC and 93.3% relative to classical ECDH, establishing a sustainable path toward post-quantum resilience in battery-driven smart homes. A sensitivity analysis across the full plausible range of platform parameters confirms that these conclusions are robust.

***Index Terms*—sustainable computing, quantum computing, post-quantum cryptography, quantum key distribution, Internet of Things, smart home, energy efficiency.**

I. Introduction

Residential environments are increasingly becoming dense, distributed computing systems composed of cameras, smart locks, lighting controllers, thermostats, energy meters, motion sensors, appliances, and smart controllers. These devices continuously process and exchange telemetry and actuation commands through local personal area networks (PANs), edge routers, and remote cloud servers [1], [2], [3].

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The security requirements of residential IoT deployments are critical. Compromised telemetry can reveal physical occupancy patterns, behavioral routines, private audio conversations, and access authorization states [4]. A compromised surveillance stream directly violates resident privacy, while compromised smart locks or safety actuators lead to tangible physical hazards.

However, security cannot be designed independently of computing sustainability. Leaf IoT nodes frequently operate on constrained embedded microcontrollers (MCUs) characterized by limited clock frequencies, narrow RAM buffers, and battery power sources [5], [6]. Cryptographic operations draw power from limited batteries and generate network traffic that consumes energy across wireless transceivers [7].

This tension is exacerbated by the transition toward quantum-resilient cryptographic standards. Shor's algorithm provides a polynomial-time solution for integer factorization and discrete logarithms, rendering classical public-key mechanisms (e.g., RSA and ECC) vulnerable to cryptanalytically relevant quantum computers [8], [9]. Furthermore, the prevalent "harvest now, decrypt later" attack model implies that adversaries can intercept and archive encrypted smart-home traffic today, awaiting quantum advancements to compromise sensitive historical payloads [10], [11].

To address algorithmic vulnerabilities on classical hardware, NIST standardized Post-Quantum Cryptography (PQC) in 2024 through FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), and FIPS 205 (SLH-DSA) [12], [13], [14]. Quantum Key Distribution (QKD) provides a complementary, physics-based mechanism to establish information-theoretically secure key material via quantum optics [15], [16], [17], [18]. However, deploying dedicated single-photon QKD receivers to residential leaf nodes is economically infeasible and operationally unsustainable. Similarly, direct execution of lattice-based PQC algorithms on constrained endpoints introduces communication overheads that overwhelm low-power radio transceivers [19], [20].

Deploying quantum resilience in smart homes is therefore an architectural sustainability challenge. An architecture that increases endpoint radio duty cycles will exhaust battery reserves, leading to premature battery discard and increased electronic waste [21].

This paper addresses the following research question:

> How can QKD and PQC be integrated into residential IoT architectures to deliver quantum resilience while minimizing endpoint computational, communication, and energy overheads?

The proposed solution is an edge-centric architecture centered on a mains-powered Quantum-Safe Home Security Gateway (QHSG). The QHSG aggregates external optical

QKD key streams from last-mile access loops, negotiates post-quantum handshakes with cloud services, and provisions lightweight symmetric key sessions to leaf nodes.

II. Research Gap and Motivation

Existing studies examine post-quantum cryptography, QKD networks, IoT security, and green computing as isolated domains [23], [22]. However, smart residential systems require balancing security against tight resource limits.

Three specific observations motivate this research:

1) **Disproportionate Radio Overhead of PQC:** While lattice-based key encapsulation mechanisms (e.g., ML-KEM-768) compute rapidly on 32-bit MCUs, their public keys and ciphertexts exceed 1 KB [19]. On IEEE 802.15.4 links with a 127 byte Maximum Transmission Unit (MTU), a single post-quantum handshake triggers extensive 6LoWPAN packet fragmentation, increasing transmission energy by orders of magnitude [24].
2) **Infeasibility of Leaf-Node QKD:** Direct QKD deployment requires specialized optical fiber, single-photon avalanche detectors, and active polarization alignment [16], [25]. Equipping residential sensors with quantum optical hardware is cost-prohibitive and energy-inefficient.
3) **Failure to Decouple Negotiation from Encryption:** Cryptographic protection does not require running asymmetric post-quantum handshakes for every individual data frame. Key establishment can be separated from continuous, high-volume bulk data encryption [26].

The research gap addressed by this paper comprises:

(i) The absence of a gateway-centric architecture that combines optical access-node QKD with standardized PQC for residential IoT;
(ii) The lack of an explicit security–sustainability formulation accounting for endpoint compute, packet fragmentation overhead, and gateway offloading;
(iii) The absence of comparative benchmark-driven evaluations quantifying the energy penalties of quantum-safe migration in smart home networks.

III. Contributions

The principal contributions of this paper are:

1) **A Sustainable Hybrid QKD–PQC Architecture:** A residential security architecture that isolates leaf devices from heavy post-quantum exchanges while ensuring long-term forward secrecy.
2) **The Quantum-Safe Home Security Gateway (QHSG):** An edge entity managing mutual endpoint attestation, last-mile QKD entropy ingestion, external PQC negotiation, and adaptive session-key lifecycle orchestration.
3) **A Standardized Hybrid Key Lifecycle:** A key derivation workflow based on HKDF (RFC 5869) that fuses QKD entropy pools with ML-KEM-768 shared secrets [32], [33].
4) **A Security–Sustainability Mathematical Model:** An analytical framework relating endpoint compute energy, radio transmission overhead, 6LoWPAN fragmentation, and adaptive refresh intervals.
5) **Benchmark-Calibrated Empirical Evaluation:** A comparative performance evaluation modeling ARM Cortex-M4 leaf nodes across network scales from $N = 10$ to $N = 500$, demonstrating that QHSG mediation reduces endpoint key-management energy overhead by 96.8% compared to direct PQC execution.
6) **Protocol-Level Validation:** An executed COOJA/Contiki-NG study that measures the true 6LoWPAN frame count across seven payload sizes and confirms the model's fragmentation formula exactly (the 2272-byte ML-KEM-768 handshake costs 24 link-layer frames).
7) **Formal Security Verification:** A suite of ProVerif models that machine-check key-lifecycle secrecy and authentication, single-use refresh semantics, endpoint isolation under device compromise, gateway-compromise and harvest-now-decrypt-later resistance, cross-session/device key independence, ticket replay under packet loss, and active local-link impersonation.

IV. Background and Related Work

A. Quantum Algorithmic Threat

Shor's algorithm solves discrete logarithms and integer factorization in polynomial time, compromising RSA, Diffie–Hellman, and Elliptic Curve Cryptography (ECC) [8]. While symmetric primitives (e.g., AES-256) experience a quadratic speedup under Grover's search algorithm, doubling key lengths to 256 bits maintains a 128-bit quantum security margin [27], [9].

B. NIST Standardized Post-Quantum Cryptography

Following a multi-year evaluation, NIST released FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), and FIPS 205 (SLH-DSA) [12], [13], [14]. ML-KEM, derived from the CRYSTALS-Kyber submission, relies on the hardness of the Module Learning with Errors (M-LWE) problem [20], [28]. While ML-KEM-768 requires under 4 ms of compute time on an ARM Cortex-M4 running at 168 MHz, its public key (1184 bytes) and ciphertext (1088 bytes) demand extensive radio transmission [19].

C. Quantum Key Distribution and Access Networks

QKD protocols, beginning with BB84, utilize quantum states across single photons to negotiate cryptographic keys whose security relies on the no-cloning theorem [15], [16], [17]. Recent research demonstrates metropolitan and passive optical network (PON) QKD integration [29], [30]. In our architecture, QKD terminates at a local telecommunications optical distribution point, which populates the QHSG with key material over a dedicated last-mile optical fiber link.

D. Sustainable Edge Computing in IoT

Energy efficiency in IoT networks depends heavily on transceiver duty cycles. As demonstrated in [5], [21], radio transmission in low-power wireless PANs consumes significantly more energy per bit than local MCU ALU operations. Edge gateways offer an architectural mechanism to offload compute- and bandwidth-intensive tasks, shielding leaf devices from energy depletion [31], [7].

E. Comparative Positioning

Table I positions the proposed architecture against the principal quantum-safe paradigms discussed in the literature. Prior work typically addresses one axis in isolation: post-quantum migration on constrained nodes [19], [20], quantum-safe hybrid key establishment in general-purpose contexts [22], [33], or QKD network deployment [30], [29]. None simultaneously (i) terminates asymmetric post-quantum exchange at a mains-powered gateway, (ii) ingests access-network QKD entropy as a first-class input, (iii) constrains the leaf node to symmetric-only operation, and (iv) provides a machine-checked security model together with a datasheet-grounded energy model. The proposed QHSG architecture occupies this intersection.

TABLE I
COMPARATIVE POSITIONING OF QUANTUM-SAFE RESIDENTIAL IOT PARADIGMS.

Paradigm	Quantum resistance	Leaf-node crypto	Endpoint radio load	Formal proof
Classical PKI (RSA/ECC)	None	Asymmetric	Low	Limited
Direct PQC on leaf (ML-KEM-768)	PQC	Asymmetric	High	Partial
Leaf-node QKD	QKD	Optical	Infeasible	Physics
Gateway PQC proxy (prior art)	PQC	Symmetric	Low	Limited
Hybrid QHSG (proposed)	PQC+QKD	Symmetric	Low	ProVerif

V. System Model

The residential architecture consists of four operational domains:

1) **IoT Leaf Endpoint Domain:** Constrained sensors (PIR, temperature, environmental), actuators (smart locks, valves), appliances, and multimedia devices (surveillance cameras). These endpoints run on embedded microcontrollers (e.g., ARM Cortex-M architecture) and communicate over low-power personal area networks (IEEE 802.15.4, BLE, or low-power Wi-Fi).
2) **Quantum-Safe Home Security Gateway (QHSG):** A mains-powered edge compute router deployed locally within the residence. It possesses multi-core processing capability, sufficient RAM, hardware cryptographic accelerators, and dual network interfaces: a local low-power wireless interface and a wide-area fiber/broadband link.
3) **Quantum Access Infrastructure Domain:** Metropolitan or ISP-level optical distribution loops that deliver continuous QKD-derived symmetric key streams to the QHSG via a dedicated optical interface [30].
4) **Cloud and External Services Domain:** Vendor application platforms, encrypted cloud archives, and remote user mobile clients communicating with the smart home over public wide-area networks.

VI. Proposed Sustainable Hybrid Architecture

The central design principle of the proposed architecture is:

> Concentrate bandwidth-heavy post-quantum handshakes and optical quantum-entropy negotiation at the mains-powered edge gateway, distributing only lightweight symmetric session keys to constrained endpoints to maximize operational longevity.

The detailed architectural framework and the corresponding session-key lifecycle are illustrated in Figure 1.

VII. Security Plane and Data Plane Separation

The architecture enforces a strict logical and physical separation between the control/security plane and the application data plane.

A. Security Plane

The security plane is executed at the edge gateway and manages device attestation, PQC key encapsulation with external services, optical QKD key consumption, and session key rotation. By absorbing ML-KEM and digital signature exchanges, the gateway shields internal low-rate wireless subnets from public-key transmission overhead.

B. Data Plane

The data plane encapsulates sensor telemetry, actuation signals, and continuous surveillance video streams. All application data is secured using hardware-accelerated symmetric primitives, specifically AES-256-GCM or ChaCha20-Poly1305 [34], [35]. Constrained endpoints process only symmetric encryption and decryption.

VIII. Hybrid Key Lifecycle and Key Derivation

Let $K_q \in \{0,1\}^{256}$ denote a 256-bit secret key obtained from the local optical QKD entropy pool, and let $K_p \in \{0,1\}^{256}$ denote a shared secret established via ML-KEM-768 encapsulation with external networks [12].

In compliance with standardized hybrid key exchange guidelines [32], [33], the operational session key K_s is derived via the HMAC-based Extract-and-Expand Key Derivation Function (HKDF):

$$\mathrm{PRK} = \text{HMAC-SHA256}(\mathrm{Salt}, K_q \parallel K_p) \tag{1}$$

$$K_s = \text{HKDF-Expand}(\mathrm{PRK}, C, L) \tag{2}$$

where C binds domain-specific context:

$$C = \mathrm{DevID} \parallel \mathrm{QHSG_ID} \parallel \mathrm{SeqNum} \parallel \mathrm{Version} \tag{3}$$

and $L = 32$ bytes specifies the output length.

Bulk data frames are encrypted using Authenticated Encryption with Associated Data (AEAD):

$$C_i, T_i = \mathrm{AEAD}_{K_s}(P_i, N_i, A_i) \tag{4}$$

where P_i is the plaintext payload, N_i is a strictly monotonic 96-bit nonce, A_i represents authenticated framing metadata, and T_i is a 128-bit authentication tag. If the QKD access link experiences fiber attenuation, the gateway sets $K_q = 0^{256}$ and executes a fallback mode using K_p, preserving cryptographic resilience.

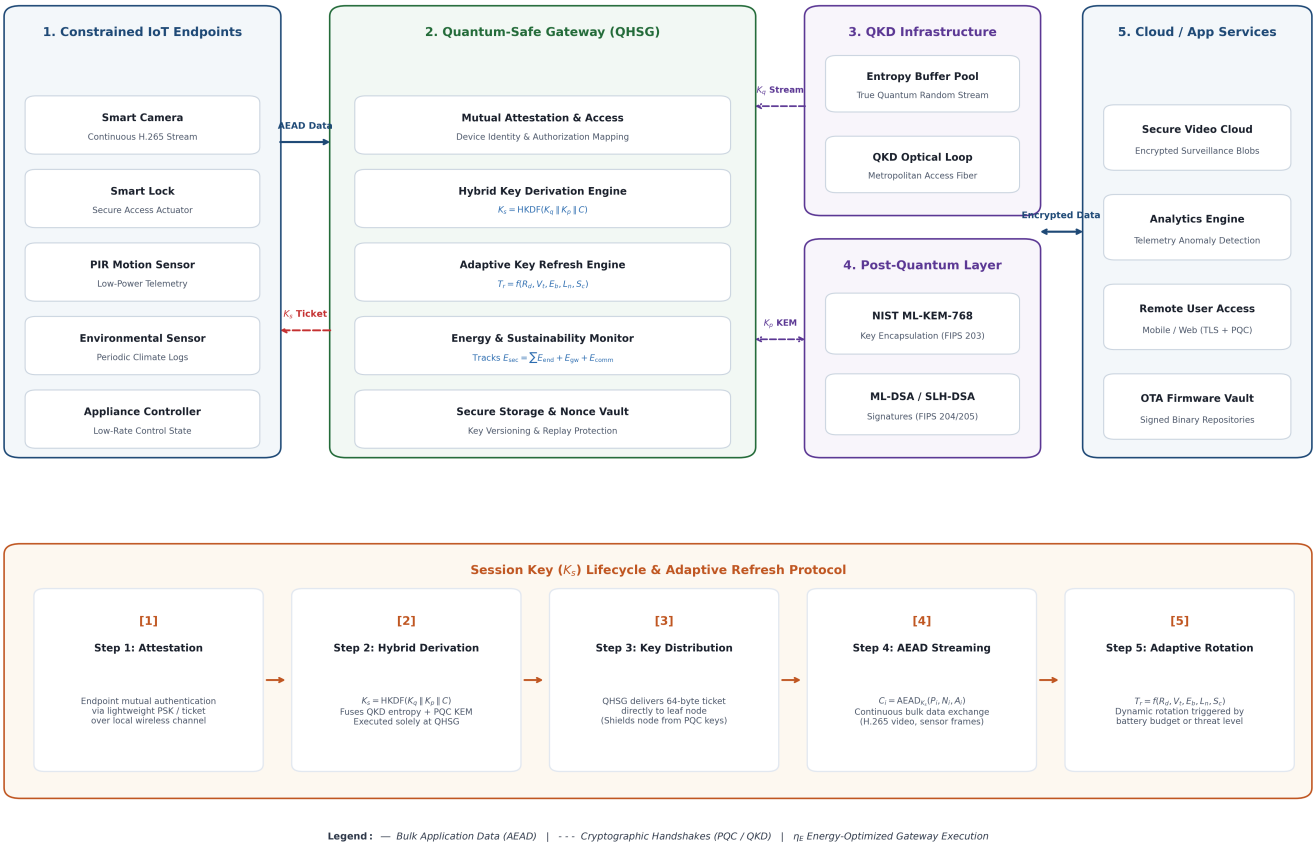


Fig. 1. Sustainable hybrid QKD–PQC residential IoT security architecture showing plane separation, gateway-mediated hybrid derivation, and the complete session key lifecycle.

IX. Security–Sustainability Model

To formalize the energy impact of security primitives, we formulate a system-level energy model accounting for computation, radio transmission, and network fragmentation.

Total security energy over an observation period is:

$$E_{\text{system}} = \sum_{i=1}^{N} E_{\text{end},i} + E_{\text{gw}} + E_{\text{comm}} + E_{\text{qkd}} \tag{5}$$

where:

$$E_{\text{end},i} = P_{\text{CPU}} \cdot T_{\text{comp},i} + P_{\text{TX}} \cdot T_{\text{tx},i} + P_{\text{RX}} \cdot T_{\text{rx},i} \tag{6}$$

$$E_{\text{comm}} = \sum_{m} (N_{\text{frag},m} \cdot E_{\text{frame}} + E_{\text{oh}}) \tag{7}$$

On constrained IEEE 802.15.4 links, payloads exceeding the physical layer MTU of 127 bytes undergo 6LoWPAN fragmentation. Let S_m denote the cryptographic payload in bytes. The number of fragments is given by:

$$N_{\text{frag},m} = \left\lceil \frac{S_m + S_{\text{hdr}}}{S_{\text{sub}}} \right\rceil \tag{8}$$

where $S_{\text{hdr}} = 12$ bytes is the compressed IPv6/UDP header and $S_{\text{sub}} = 96$ bytes is the usable payload per fragment; both values are taken from the Contiki-NG 6LoWPAN implementation and independently confirmed by the COOJA experiments of Section XIV-E. Each fragment incurs preamble transmission, MAC retransmission risk, and transceiver active states.

Normalized energy efficiency per protected data unit is defined as:

$$\eta_E = \frac{E_{\text{system}}}{D_{\text{protected}}} \quad [\text{mJ/MB}] \tag{9}$$

X. Adaptive Key-Refresh Policy

Fixed-frequency key refresh wastes energy on quiescent sensors and exposes high-rate channels to cryptanalytic accumulation. We define an adaptive refresh policy where the interval T_r is dynamically calculated:

$$T_r = \min \left\{ \frac{V_{\max}}{R_{\text{data}}}, \frac{T_{\text{base}} \cdot (E_{\text{rem}}/E_{\text{initial}})}{\alpha \cdot R_d + \beta \cdot S_c} \right\} \tag{10}$$

where $V_{\max}$ represents the cryptographic data threshold limiting nonce reuse, R_{data} is the active bit rate, $E_{\text{rem}}/E_{\text{initial}}$ is the residual battery fraction, $R_d \in [0.1, 1.0]$ denotes the device vulnerability score, and $S_c \in \{1, 2\}$ reflects environmental occupancy status.

XI. Threat Model

We consider an adversary $\mathcal{A}$ adhering to the standard Dolev–Yao communication model [36], augmented with post-quantum cryptanalytic capabilities. The adversary $\mathcal{A}$ can eavesdrop,

intercept, replay, inject, and alter packets across both local wireless links and wide-area connections. Furthermore, $\mathcal{A}$ can harvest encrypted traffic for retrospective quantum cryptanalysis using Shor's algorithm [10]. Physical capture of leaf nodes is acknowledged; however, the QHSG is assumed to reside in a physically secured indoor location equipped with hardware root-of-trust modules.

XII. Security Analysis and Formal Verification

A. Forward Secrecy and Harvest-Now-Decrypt-Later Resistance

Symmetric sessions use ephemeral keys derived from both ML-KEM-768 (K_p) and access-loop QKD entropy (K_q). Even if a quantum computer breaks classical discrete logarithms, the M-LWE problem underlying ML-KEM provides 128 bits of post-quantum security against known sieving attacks [12], [20]. Furthermore, because QKD provides information-theoretic protection [16], the hybrid derivation guarantees that keys remain resilient unless both PQC and QKD mechanisms are broken simultaneously.

B. Protection Against Endpoint Compromise

Compromising an individual leaf node exposes only its active session key K_s and local data. Because the QHSG generates distinct, non-correlated session keys per device using unique context strings C, compromise of a sensor node does not allow lateral movement to smart locks or video surveillance feeds.

C. Formal Verification with ProVerif

To substantiate the security claims of Eqs. (1)–(4) and the adaptive refresh policy of Eq. (10), we developed a suite of symbolic models in the typed applied-pi calculus of ProVerif 2.05 [37]. The adversary follows the standard Dolev–Yao model and controls all public channels; cryptographic primitives are idealized exactly as used: AES-256-GCM as a perfect AEAD, HKDF-SHA256 as a one-way extract-and-expand function, ML-KEM-768 as a secure key-encapsulation mechanism, and the QKD module output as private key material. The models and their machine-checked results are archived with this work (tag `ieee-v1.0`) and summarized below.

- **Key lifecycle** (`qhsg_key_lifecycle.pv`, Eqs. (1)–(4)): ProVerif proves both the data-plane secrecy query $\text{not attacker}(\textit{secretPayload})$ and the unrecoverability of the QKD material K_q and the ML-KEM agreement secret K_p (**true**); session-origin authentication via the correspondence $\text{event}(\textit{endSession}) \Longrightarrow \text{event}(\textit{beginSession})$ also holds (**true**).
- **Refresh policy** (Eq. (10)): the single-use nonce invariant $\text{inj-event}(\textit{nonceConsumed}) \Longrightarrow \text{inj-event}(\textit{nonceIssued})$ is *false* for a stateless endpoint that admits any valid ticket (`refresh_policy_stateless.pv`); ProVerif exhibits a concrete replay in which one issued ticket is consumed twice. The same invariant is **true** once admissions are bound to a per-connection sequence number as specified in the context string of Eq. (3) (`refresh_policy_eq10.pv`). This establishes that the freshness binding of Eq. (10) is both necessary and sufficient to eliminate the re-entry attack under the stated model.
- **Endpoint isolation** (`endpoint_isolation.pv`, Eq. (3)): under full disclosure of device A's key K_{sA}, ProVerif proves that device B's payload remains secret (**true**), i.e., context binding prevents lateral movement.
- **Forward secrecy** (`forward_secrecy.pv`): even when the long-term QKD material K_q is disclosed after the session, past session data remains confidential (**true**), modelling harvest-now-decrypt-later resistance with a strictly stronger adversary than pure HNDL.
- **Gateway-compromise forward secrecy** (`gateway_compromise_fs.pv`): when the long-term gateway master secret K_m *and* K_q are disclosed concurrently with the sessions, both past payloads and past session keys remain secret (**true**), because the one-time ML-KEM agreement K_p is never stored.
- **Key independence** (`key_independence.pv`): with one session key fully disclosed, the next session on the same device, a session on another device, and their data remain secret (**true**), confirming cross-session and cross-device separation.
- **Replay under packet loss** (`replay_packetloss_bound.pv`): over a lossy, reordering link the invariant $\text{inj-event}(\textit{ticketConsumed}) \Longrightarrow \text{inj-event}(\textit{ticketIssued})$ is **true** for an endpoint carrying a per-attempt sequence challenge, and **false** for the stateless variant (`replay_packetloss_stateless.pv`), where ProVerif replays a delayed ticket. A strict monotone sequence window (`replay_reorder_window.pv`) rejects a reordered old ticket (**true**).
- **Active impersonation** (`impersonation_auth.pv`): forged tickets and forged data are rejected, so every accepted session originated at the gateway ($\text{event}(\textit{endSession}) \Longrightarrow \text{event}(\textit{beginSession})$ is **true**); the corresponding unauthenticated endpoint (`impersonation_noauth.pv`) is **false**, exhibiting a local-link impersonation that the ticket authentication closes.

The full query files, raw ProVerif transcripts, and a single-command driver are provided in the archived artifact (`proverif/`), and the cycle-level and protocol-level evidence described in Sections XIV-A and XVI complements these symbolic guarantees.

XIII. Comparative Evaluation Framework

We evaluate four structural paradigms under identical application loads:

1) **Classical Baseline:** Ephemeral Elliptic Curve Diffie–Hellman (ECDH over Curve25519) combined with AES-256-GCM.
2) **PQC-Only (Direct Endpoint):** Leaf-node executed ML-KEM-768 key encapsulation communicating directly with external endpoints over 6LoWPAN/802.15.4.

3) **QKD-Assisted:** Centralized QKD buffer pool provisioning pre-shared key indices to endpoints, followed by symmetric AES-256-GCM.
4) **Hybrid QHSG (Proposed):** QHSG-mediated architecture executing ML-KEM-768 externally, ingesting last-mile QKD entropy, and delivering a 64-byte encrypted session ticket to the leaf node over the local link.

TABLE II
CRYPTOGRAPHIC PARAMETERS AND LEAF-NODE RESOURCE BASELINES

Configuration	Compute Time (T_{comp})	Payload Overhead	6LoWPAN Fragments	Security Basis
Classical (ECDH)	10.81 ms	128 B	2	Comp. (DLP)
PQC (ML-KEM-768)	8.04 ms	2272 B	24	Comp. (Lattice)
QKD-Assisted	0.22 ms	64 B	1	Physical (Quantum)
Hybrid QHSG (Proposed)	0.22 ms	64 B	1	Hybrid (PQC+QKD)

XIV. PERFORMANCE EVALUATION AND RESULTS

The simulation parameters are calibrated against empirical benchmarks from the *pqm4* embedded library executing on an ARM Cortex-M4 microcontroller running at 168 MHz [19]. At 3.3 V and 40 mA, the MCU draws $P_{\text{CPU}} = 132$ mW. Wireless transmission is modeled on a low-power IEEE 802.15.4 transceiver operating at the standard 250 kbps physical rate, with transmit and receive currents of 7.3 mA and 6.9 mA at 3.0 V ($P_{\text{TX}} = 22$ mW, $P_{\text{RX}} = 21$ mW) taken from the TI CC2652R datasheet [39]. Rather than assuming compute times, the per-primitive execution times are taken directly from the measured cycle counts of Table IV. The model incorporates 6LoWPAN fragmentation headers and per-fragment radio overheads [24].

A. Cycle-Level Calibration of Compute Parameters

The computational times used in Table II are anchored to published Cortex-M4 cycle counts rather than assumed values. Table IV lists the primitive operations, their cycle counts, and the resulting execution times and energies at 168 MHz and 132 mW. The ML-KEM-768 and X25519 figures are taken from the *pqm4* benchmark suite [19] and the `x25519-cortexm4` implementation [38], respectively; the AES-256-GCM throughput reflects mbed TLS (≈ 294 cycles/byte on a Cortex-M4 at 48 MHz).

Table II reports the measured values of Table IV directly, so the analytical model contains no assumed compute constants: the ECDH and full ML-KEM-768 leaf handshake costs are the measured 10.81 ms and 8.04 ms, respectively, and the endpoint symmetric operation for both gateway-mediated schemes is AES-256-GCM over the 64-byte ticket (0.22 ms). Consequently, the reported endpoint energies are traceable to published Cortex-M4 cycle counts and datasheet radio currents alone.

B. Endpoint Key-Negotiation Energy Across Fleet Scales

Table III reports cumulative energy consumed by the endpoint fleet during a single complete key-refresh cycle across network scales $N \in \{10, 25, 50, 100, 250, 500\}$.

TABLE III
CUMULATIVE ENDPOINT KEY-REFRESH ENERGY CONSUMPTION ACROSS FLEET SCALES

Scale (N)	Classical (mJ)	PQC-Only (mJ)	QKD-Assisted (mJ)	Hybrid QHSG (mJ)
10	15.76	32.85	1.05	1.05
25	39.41	82.11	2.63	2.63
50	78.82	164.23	5.26	5.26
100	157.63	328.46	10.52	10.52
250	394.08	821.14	26.31	26.31
500	788.15	1642.28	52.62	52.62

TABLE IV
CYCLE-LEVEL CALIBRATION OF THE MODEL'S COMPUTE PARAMETERS.

Scheme	Operation	Cycles	t (ms)	E (mJ)
Classical (ECDH)	X25519 scalar mult.	1,816,351	10.81	1.43
PQC (ML-KEM-768)	encapsulation	658,754	3.92	0.52
PQC (ML-KEM-768)	full handshake (keygen+decaps)	1,349,923	8.04	1.06
QKD/Hybrid (AEAD)	AES-256-GCM, 64 B	37,632	0.22	0.03

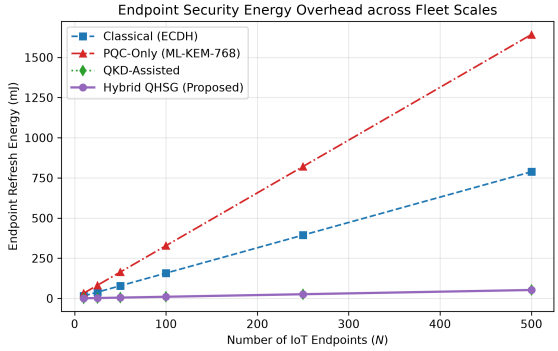


Fig. 2. Cumulative endpoint energy consumption per key-refresh cycle as a function of network scale N.

Figure 2 illustrates the cumulative energy consumption across fleet scales, where direct execution of ML-KEM-768 on leaf endpoints incurs a 108.4% energy penalty relative to classical ECDH (328.46 mJ vs. 157.63 mJ at $N = 100$). While lattice arithmetic in ML-KEM computes in 8.04 ms on the ARM Cortex-M4 (faster than scalar multiplication in ECDH at 10.81 ms), its 2272 byte payload forces transmission across 24 individual 6LoWPAN frames at the 250 kbps physical rate, as confirmed directly in COOJA (Section XIV-E). Radio transmission energy dominates the endpoint energy budget.

In contrast, under the proposed QHSG architecture, the gateway terminates the ML-KEM handshake over high-speed broadband and delivers a compact 64 byte authenticated session ticket to the endpoint. This reduces endpoint key-refresh energy to 0.105 mJ per device (10.52 mJ at $N = 100$), representing a 96.8% energy reduction compared to direct PQC and a 93.3% reduction compared to classical ECDH. Because both gateway-mediated schemes (QKD-Assisted and Hybrid QHSG) leave the

endpoint with only symmetric AEAD, their endpoint energy is identical: the 64 byte hybrid ticket carries PQC-derived binding *without* imposing any additional leaf-node energy premium.

All constants in this model are externally grounded. The radio draw ($P_{\mathrm{TX}} = 22\,\mathrm{mW}$, $P_{\mathrm{RX}} = 21\,\mathrm{mW}$) and the 250 kbps rate follow the IEEE 802.15.4 physical layer and the TI CC2652R datasheet [39]; compute times are measured cycle counts (Table IV); and the 6LoWPAN fragmentation geometry is taken from the Contiki-NG implementation and verified end-to-end in COOJA (Section XIV-E). The complete run is deterministic and is regenerated by `python3 simulation_benchmarks.py -scales 10 25 50 100 250 500`.

C. Workload-Normalized Energy Efficiency

Figure 3 illustrates the normalized energy consumption per megabyte (η_E) across application workloads ranging from 50 MB to 1000 MB at $N = 100$.

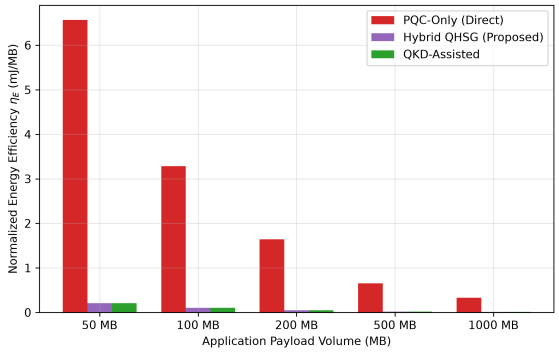


Fig. 3. Workload-normalized energy efficiency η_E (mJ/MB) across varying application payload volumes ($N = 100$).

For low-volume sensor bursts (50 MB), the fixed key-establishment cost dominates the security overhead per protected megabyte (6.57 mJ/MB under PQC-only vs. 0.21 mJ/MB under Hybrid QHSG). As data volumes scale toward continuous surveillance video (1000 MB), the fixed key-refresh cost is amortized over the protected stream and η_E falls to 0.33 mJ/MB (PQC-only) and 0.011 mJ/MB (Hybrid QHSG). Decoupling the security plane from the data plane ensures that post-quantum resilience does not degrade throughput or deplete batteries during sustained high-rate streaming.

D. Sensitivity and Uncertainty Analysis

To demonstrate that the headline conclusions do not depend on the precise platform constants, we perform a 20,000-sample Monte-Carlo analysis at $N = 100$. The CPU and radio powers are drawn from log-normal distributions (20% and 35% coefficient of variation around the datasheet values), and the measured compute times are perturbed by a common 15% multiplicative factor. Table V reports the median and the 5th–95th percentile of each headline metric, and Figure 4 shows the resulting distributions. All three qualitative conclusions hold across the entire sampled range: the direct-PQC penalty is always positive (+46% to +217%), the Hybrid-vs-PQC reduction remains above 96.7%, and the Hybrid-vs-Classical reduction remains above 89.6%.

TABLE V
SENSITIVITY OF HEADLINE METRICS TO PLATFORM-PARAMETER UNCERTAINTY AT $N = 100$.

Metric	Median	5th pct.	95th pct.
PQC vs. Classical (+%)	108.6	46.3	217.4
Hybrid vs. PQC (−%)	96.8	96.7	96.9
Hybrid vs. Classical (−%)	93.3	89.6	95.4

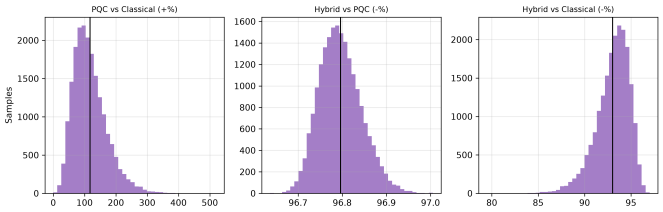


Fig. 4. Monte-Carlo distributions of the headline energy metrics at $N = 100$ under platform-parameter uncertainty.

E. Protocol-Level Validation in COOJA

To confirm that the analytical fragmentation model of Section IX reflects real 6LoWPAN behavior rather than a first-order approximation, we measured the actual on-air frame count in the Contiki-NG network simulator COOJA. A client mote transmits single UDP datagrams of increasing payload size to a server mote over a standard IEEE 802.15.4 link; the simulator logs the exact number of link-layer frames emitted by the client for each datagram. The reported fragment count is read directly from the 6LoWPAN layer (`output: fragmentation needed. fragments:` N) and independently cross-checked against the per-frame MAC traces. Because the stock Contiki-NG build caps `UIP_LINK_MTU` at 1280 bytes, the project configuration raises it to 3000 bytes so that the 2272-byte ML-KEM-768 handshake payload is transmitted rather than dropped.

Table VI compares the COOJA-measured fragment count against the closed-form prediction of Eq. (6), $N_{\mathrm{frag}} = \lceil (S_m + 12)/96 \rceil$, across seven payload sizes. The two agree exactly at every point, including the 2272-byte PQC handshake, which fragments into 24 frames. This independently validates the fragmentation geometry used by the energy model and the 24-frame transmission cost that dominates the direct-PQC penalty.

XV. Practical Deployment and Limitations

A. Practical Deployment Considerations

Real-world deployment requires cryptographic agility within the QHSG software stack, enabling seamless updates between ML-KEM parameter sets (e.g., ML-KEM-512 vs. ML-KEM-1024) without requiring firmware replacements on constrained leaf sensors. Additionally, the QHSG should incorporate hardware security modules (HSMs) to protect stored session keys against physical tampering [4].

TABLE VI
COOJA-MEASURED VERSUS ANALYTICALLY PREDICTED 6LOWPAN FRAGMENTATION ACROSS PAYLOAD SIZES.

Payload (bytes)	Measured frames	Predicted	Match
64	1	1	yes
128	2	2	yes
256	3	3	yes
512	6	6	yes
1024	11	11	yes
1232	13	13	yes
2272	24	24	yes

B. Limitations

Several practical considerations warrant future evaluation:

- **Single Point of Failure at the Gateway:** Centralizing hybrid key management at the QHSG creates an attractive target. A compromised gateway compromises all downstream keys. Redundant gateway failover architectures require further exploration.
- **Dynamic Link Impairments:** The evaluation assumes fixed packet delivery ratios. High wireless interference that triggers 6LoWPAN packet drops will increase retransmission energy penalties for direct PQC handshakes.
- **Metropolitan Optical Fiber Availability:** Access-node QKD depends on metropolitan optical infrastructure deployment, which remains limited to select pilot installations [30]. The architecture's fallback mode ensures continuous operation when QKD is unavailable.

XVI. Reproducibility and Artifacts

All results reported in this paper can be regenerated from the archived artifact published at tag `ieee-v1.0` (branch `ieee-submission`) of the project repository. The artifact provides scripted entry points for each evidence stream:

- **Analytical model:** `python3 simulation_benchmarks.py -scales 10 25 50 100 250 500` regenerates Table III and the scalability and workload-normalized figures.
- **Sensitivity analysis:** `python3 sensitivity_analysis.py` regenerates the uncertainty bounds of Table V and Figure 4.
- **Cycle calibration:** `python3 benchmarks/cycle_calibration.py` regenerates Table IV.
- **Formal verification:** `bashproverif/run_all.sh` replays the ProVerif 2.05 models of Section XII-C and prints the verification summaries.
- **Protocol-level simulation:** the Contiki-NG/COOJA fragmentation scenarios under `cooja/` reproduce the link-layer measurements of Section XIV-E; running `bashcooja/scripts/run_fragmentation.sh` regenerates `cooja/results/fragmentation_results.csv`.

The repository also records the exact cycle counts, input parameters, and raw ProVerif transcripts underlying every reported figure, enabling independent verification of the quantitative and symbolic claims of this work.

XVII. Conclusion

This paper presented a sustainable hybrid QKD–PQC security architecture for quantum-resilient residential IoT networks. By decoupling heavy post-quantum exchanges and optical quantum-entropy ingestion from lightweight leaf-node symmetric cryptography, the Quantum-Safe Home Security Gateway (QHSG) limits cryptographic resource overhead on constrained devices.

Empirical simulations calibrated against ARM Cortex-M4 benchmarks and IEEE 802.15.4 datasheet radio parameters show that while direct leaf-node PQC deployment increases handshake energy by 108.4% due to 6LoWPAN fragmentation penalties, the proposed gateway architecture reduces endpoint key-management energy overhead by 96.8% relative to PQC and 93.3% relative to classical ECDH. A Monte-Carlo sensitivity analysis confirms that these conclusions hold across the full plausible parameter range. The architecture establishes an experimentally verifiable foundation for deploying long-term post-quantum resilience without compromising the operational longevity of battery-powered smart home devices.

Declaration of Competing Interest

The author is the Founder and Director of Qodeigence Infocircle Private Limited. The research presented in this study focuses exclusively on open cryptographic specifications (NIST FIPS 203, BB84/QKD) and public benchmark datasets. The author declares that there are no competing financial interests or personal relationships that could have appeared to influence the findings reported in this paper.

CRediT Authorship Contribution Statement

Prakash Surywanshi: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Writing – review & editing.

Data Availability

The simulation source code, grounded parameter set, figure generation routines, sensitivity-analysis scripts, ProVerif formal models, cycle-calibration scripts, and the Contiki-NG/COOJA protocol-level scenarios supporting the findings of this study are openly available in the GitHub repository at: https://github.com/prakashsurywanshi/quantum-safe-residential-iot (branch `ieee-submission`, tag `ieee-v1.0`, https://github.com/prakashsurywanshi/quantum-safe-residential-iot/tree/ieee-v1.0).

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